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INSTITUTE OF BIOINFORMATICS AND APPLIED BIOTECHNOLOGY

Term End Examination - **TEE | MSc (2024-2026) Second Semester** | 21 May 2025

BDBP207: MACHINE LEARNING LABORATORY

Time: 240 minutes *Maximum Marks: 70 (60+10)*

PART A	Model Development(answer any 1 out of 2)	(40 Marks)

A1. The OJ dataset (Orange Juice dataset) from the *Introduction to Statistical Learning with Python (ISLP)* package comprises 1,070 observations of customer purchases between two orange juice brands: Citrus Hill (CH) and Minute Maid (MM). Each entry captures various attributes related to the purchase, including pricing, promotions, customer loyalty, and store information. Use this dataset for solving this problem. The dataset features are:

1. **Purchase**: Indicates the brand purchased (CH or MM).
2. **WeekofPurchase**: Week number when the purchase occurred.
3. **StoreID**: Identifier for the store.
4. **PriceCH:** Regular prices for Citrus Hill
5. **PriceMM**: Regular prices for Minute Maid
6. **DiscCH:** Discounts offered on CH
7. **DiscMM**: Discounts offered on MM.
8. **SpecialCH:** Indicators if CH is on special promotion.
9. **SpecialMM**: Indicators if MM is on special promotion.
10. **LoyalCH**: Customer's loyalty score towards CH.
11. **SalePriceCH:** Effective sale prices after discounts.
12. **SalePriceMM**: Effective sale prices after discounts.
13. **PriceDiff**: Difference between MM and CH sale prices.
14. **Store7**: Indicates if the purchase was made at Store 7(Yes or No).
15. **PctDiscCH:** Percentage discounts for CH
16. **PctDiscMM**: Percentage discounts for MM.
17. **ListPriceDiff**: Difference in list prices between MM and CH.
18. **STORE**: Categorical variable indicating one of five possible stores.

(a) Create a training set containing a random sample of 800 observations, and a test set containing the remaining observations.

(b) Fit a support vector classifier (svm.LinearSVC) to the training data using C = 0.01, with Purchase as the target and the other variables as features. Measure the train and test accuracies.

(c) Use GridSearchCV to select an optimal C. Consider values 0.01, 1, 10.

(d) Compute the training and test error rates using this new value for C.

(e) Fit a support vector machine with a radial kernel. Use the default value for gamma and measure the train and test accuracies.

Overall, which approach seems to give the best results on this data?

A2. Use the NCI160 data for solving this problem. It consists of gene expression levels for 64 cancer cell lines (not just 60) measured on 6830 genes. Each observation (cell line) is labeled with a cancer type (e.g., breast, lung, colon, leukemia, etc.). Perform hierarchical clustering on the cell lines.

(a) Using hierarchical clustering with complete linkage and Euclidean distance, cluster the cell lines.

(b) Cut the dendrogram at a height that results in four distinct clusters. Which cell lines belong to which clusters?

(c) Using K-Means algorithm, perform clustering on the same dataset with k = 4.

(d) Are the four clusters obtained from Hierarchical clustering and K-Means clustering the same? Discuss and provide your insights as a comment in the code.

PART B	Model Design Considerations (answer any 1 out of 2)	(20 Marks)

B1. This question should be answered using the Weekly data set, which is part of the ISLP package. The Weekly dataset comprises 1,089 weekly stock market returns for 21 years. Implement the Leave-One-Out-Cross-Validation (LOOCV) algorithm from scratch.

(a) Write a for loop from i = 1 to i = n, where n is the number of observations in the data set, that performs the following step: Fit a logistic regression model using all but the ith observation to predict Direction using Lag1 and Lag2. The 'i'th observation is the test data and the remaining 'n-1' observations are the training data.

(b) Take the average of the n numbers obtained in (a) in order to obtain the LOOCV estimate for the test accuracy. Print the estimated average test accuracy.

B2. Let $x = [x_1, x_2, x_3]$, and $y = [y_1, y_2, y_3]$. Implement a feature mapping function, $\phi(x)$, called Transform(x), given by $\phi(x) = (x_1x_1, x_1x_2, x_1x_3, x_2x_1, x_2x_2, x_2x_3, x_3x_1, x_3x_2, x_3x_3)$.

(a) Let x = [1, 2, 3], y = [4, 5, 6]. Use the above "Transform" function to transform these vectors to a higher dimension and compute the dot product in a higher dimension. Print the value.

(b) Implement a kernel, $K(x,y) = (< x,y >)^2$. $<x,y>$ is a dot product of x and y. Apply this kernel function and evaluate the output for the same x and y vectors. Show that the result is the same in both scenarios demonstrating the power of a kernel trick.

PART C	Viva Voce	(10 Marks)